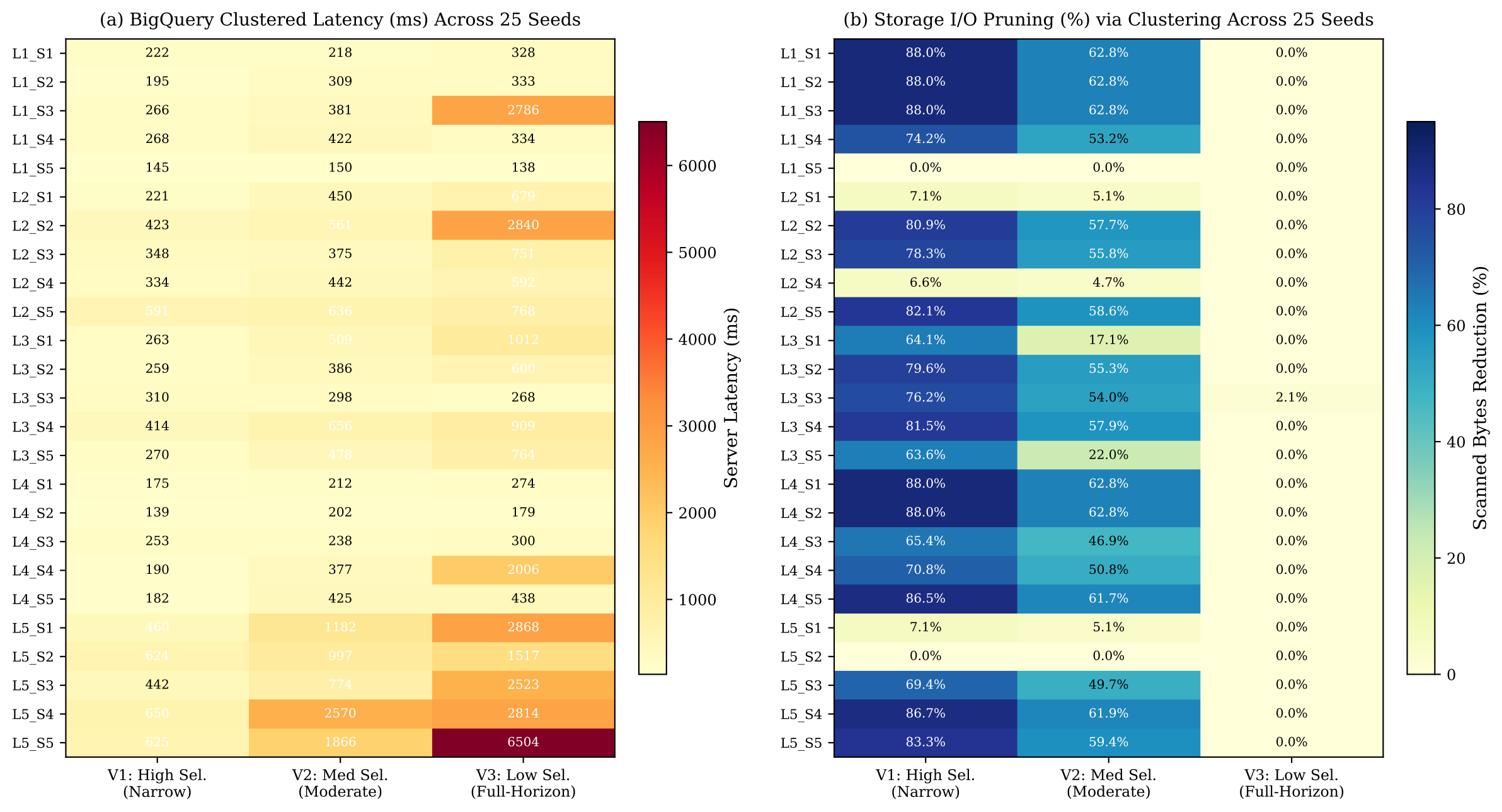

(a) BigQuery Clustered Latency (ms) Across 25 Seeds
L1_S1 222 218 328
L1_S2 195 309 333
L1_S3 266 381 2786
L1_S4 268 422 334
L1_S5 145 150 138
L2_S1 221 450 679
L2_S2 423 561 2840
L2_S3 348 375 751
L2_S4 334 442 592
L2_S5 591 636 768
L3_S1 263 509 1012
L3_S2 259 386 600
L3_S3 310 298 268
L3_S4 414 656 909
L3_S5 270 478 764
L4_S1 175 212 274
L4_S2 139 202 179
L4_S3 253 238 300
L4_S4 190 377 2006
L4_S5 182 425 438
L5_S1 460 1182 2868
L5_S2 624 997 1517
L5_S3 442 774 2523
L5_S4 650 2570 2814
L5_S5 625 1866 6504
V1: High Sel. (Narrow)
V2: Med Sel. (Moderate)
V3: Low Sel. (Full-Horizon)
Server Latency (ms)
1000
2000
3000
4000
5000
6000
(b) Storage I/O Pruning (%) via Clustering Across 25 Seeds
L1_S1 88.0% 62.8% 0.0%
L1_S2 88.0% 62.8% 0.0%
L1_S3 88.0% 62.8% 0.0%
L1_S4 74.2% 53.2% 0.0%
L1_S5 0.0% 0.0% 0.0%
L2_S1 7.1% 5.1% 0.0%
L2_S2 80.9% 57.7% 0.0%
L2_S3 78.3% 55.8% 0.0%
L2_S4 6.6% 4.7% 0.0%
L2_S5 82.1% 58.6% 0.0%
L3_S1 64.1% 17.1% 0.0%
L3_S2 79.6% 55.3% 0.0%
L3_S3 76.2% 54.0% 2.1%
L3_S4 81.5% 57.9% 0.0%
L3_S5 63.6% 22.0% 0.0%
L4_S1 88.0% 62.8% 0.0%
L4_S2 88.0% 62.8% 0.0%
L4_S3 65.4% 46.9% 0.0%
L4_S4 70.8% 50.8% 0.0%
L4_S5 86.5% 61.7% 0.0%
L5_S1 7.1% 5.1% 0.0%
L5_S2 0.0% 0.0% 0.0%
L5_S3 69.4% 49.7% 0.0%
L5_S4 86.7% 61.9% 0.0%
L5_S5 83.3% 59.4% 0.0%
V1: High Sel. (Narrow)
V2: Med Sel. (Moderate)
V3: Low Sel. (Full-Horizon)
Scanned Bytes Reduction (%)
0
20
40
60
80